Requirement Analysis and Feasibility Study

POWER WORLD GYM - KIRIBATHGODA

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Introduction

The proposed Gym Management System is a comprehensive digital solution designed to replace the existing manual and semi-digital processes at Power World Gym, Kiribathgoda. The current system, reliant on physical cards, paper-based logs, and static schedules, is inefficient, prone to error, and limits member engagement and administrative oversight.

This new integrated system aims to streamline all core gym operations into a single, secure, and user-friendly platform. Its primary objective is to automate key processes such as member access, attendance tracking, subscription management, and appointment scheduling. By digitizing these workflows, the system enhances operational accuracy, improves security, enriches the member experience through self-service features and progress tracking, and provides management with valuable, data-driven insights for strategic decision-making. The system is designed to be scalable, reliable, and accessible via various endpoints like mobile devices, and web portals.

Proposed System Functions Overview

The core functions of the proposed system, as derived from the use cases, are categorized as follows:

A. Member Access & Attendance Management:

- QR-Based Access Control (UC-01): Members scan a secure QR code at the entrance for identity verification, automated subscription validation, and door unlocking.
- Automated Subscription Validation (UC-02): A backend service that checks membership status and entitlements in real-time upon access attempts or booking requests.
- Authentication Service (UC-03): Verifies user identity during the check-in/out process.
- Timestamp Logging (UC-04): Precisely records all member arrivals and departures, preventing duplicates and ensuring accurate attendance data.
- Staff Access (UC-09): Manages secure entry and exit for staff members, integrating with the same logging system.

B. Membership & Subscription Management:

- Subscription Purchase & Renewal (UC-05): Allows members to seamlessly buy or renew subscriptions online via integrated payment gateways, with automatic activation and confirmation.

C. Scheduling & Appointment Management:

- Trainer Appointments (UC-06): Enables members to view instructor availability, book sessions, and receive automated confirmations and reminders, with checks for subscription eligibility.
- Workout Schedule Modification (UC-10): Allows members and instructors to edit workout plans, with system checks for conflicts and trainer availability.

D. Member Self-Service & Engagement:

- Workout Completion Tracking (UC-07): Allows members to mark workouts as complete, triggering progress updates.
- Progress Monitoring & Analytics (UC-08): Provides members with a dashboard displaying their workout history, trends, consistency metrics, and personal achievements.

E. Administrative & Management Functions:

- Insights Dashboard (UC-11): Provides instructors and managers with high-level analytics on gym trends, occupancy patterns, and recommended actions.
- Detailed Reporting (UC-12): Allows authorized managers to generate and export custom reports on attendance, revenue, and facility utilization.
- System Administration (UC-13): A comprehensive console for administrators to manage all system aspects, including user roles, pricing plans, facility configurations, and hardware settings.

Business System Options Comparison

BSO 1: Essential Digitalization & Access Control

- Description: This option addresses the most critical flaw: the disconnect between online payments and gym access. It implements a mandatory QR-Based Access Control (UC-01-04) system. This forces digital attendance logging and, crucially, validates the online subscription (UC-02, UC-05) at the door before granting entry. It replaces the physical sign-in book and ensures only paying members enter.
- Solves Current Pain Points:
    - Eliminates the risk of members entering with expired/without subscriptions.
    - Automates and mandates attendance tracking, replacing the optional paper log.
    - Provides a definitive record of member visits.
- Pros:
    - Highest Impact for Cost: Directly stops revenue leakage.
    - Faster Member Entry: Scans are quicker than manual sign-in.
    - Foundation for Future Systems: Establishes the core digital member database.
- Cons:
    - Requires investment in hardware (QR scanners/tablets for entrance).
    - Members must adapt to using phones/QR codes instead of just walking in.

BSO 2: Operational Efficiency & Member Portal

- Description: This option builds on BSO 1 by digitizing the other major paper-based processes. It adds the Member Self-Service Portal featuring Trainer Appointments (UC-06) and Digital Workout Scheduling (UC-10). This eliminates

the outdated "generic workout card" and allows members to book and manage their routines via an app, reducing constant instructor admin tasks.

- Solves Current Pain Points:
    - Replaces physical workout cards with dynamic, updatable digital plans.
    - Streamlines the ad-hoc booking process, reducing scheduling conflicts and instructor workload.
    - Provides a clear value proposition to members (convenience), improving retention.
- Pros:
    - Significantly reduces administrative burden on instructors.
    - Improves member satisfaction and engagement through modern self-service tools.
    - Creates a new structured data source (appointment history).
- Cons:
    - Higher development cost than BSO 1.

BSO 3: Integrated Growth & Insights Platform (Recommended)

- Description: This is the complete system as proposed. It includes BSO 1 and 2 and adds the data analytics layer. This introduces Progress Tracking for members (UC-07, UC-08) and Insights & Reporting for management (UC-11, UC-12, UC-13). This turns operational data into strategic assets.
- Solves Current Pain Points:
    - Replaces the non-existent "Manual Vitals Input" with a seamless digital progress journey, a powerful member retention tool.
    - Empowers "Top Management" and "Admin" with real-time insights into attendance trends, revenue, and facility usage, moving them from passive overseers to active managers.
    - Provides a complete 360° view of both business health and member engagement.

- Pros:
    - Maximizes ROI: Uses data to drive membership retention, personalized marketing, and operational decisions (e.g., staff scheduling during peak hours).
    - Creates a Competitive Advantage: Offers features (progress analytics, insights) that differentiate the gym from competitors.
    - Future-Proofs the Business: The integrated platform can easily scale to add new services.
- Cons:
    - Highest initial investment and complexity.
    - Requires the most cultural shift in how both staff and members interact with the gym.

Conclusion and Recommendation

- BSO 1 is a pragmatic first step focused purely on securing revenue and core process digitization.
- BSO 2 is a significant upgrade aimed at boosting operational efficiency and member satisfaction.
- BSO 3 is the strategic choice for long-term growth and market leadership, transforming the gym from a traditional facility into a data-informed, modern fitness service.

Feature Category	**Specific Feature / Use Case**	**BSO 1: Essential Digitalization & Access Control**	**BSO 2: Operational Efficiency & Member Portal**	**BSO 3: Integrated Growth & Insights Platform**
Core Access & Security	UC-01: Scan QR Code (Entry/Exit)	X	X	X
	UC-02: Validate Subscription	X	X	X
	UC-03: Authenticate User	X	X	X
	UC-04: Record Timestamp (Auto Attendance)	X	X	X
	UC-09: Staff Enter/Leave Gym	X	X	X
Membership Management	UC-05: Buy/Renew Subscription Online	X	X	X
Scheduling & Appointments	UC-06: Make a Trainer Appointment		X	X
	UC-10: Change/Modify Workout Schedule		X	X
Member Engagement & Tracking	UC-07: Mark Workout Done		X	X

	UC-08: View/Track Progress (Dashboard)		X	X
Business Intelligence & Admin	UC-11: View Insights (Analytics)			X
	UC-12: Access Detailed Reports			X
	UC-13: Manage Entire System (Admin Console)			X

Requirements analysis – Functional

Functional requirements specify the core operations and features a system must deliver to meet its intended purpose. They define what the system is expected to do and outline the services it should provide to users.

These requirements typically address:

1. System Capabilities – The essential functions and tools the system must support.
2. User Interactions – The ways in which users engage with the system, including inputs, outputs, and interface behavior.
3. Data Management – How the system processes, stores, retrieves, and secures information.

ID	**Category**	**Requirement Description**	**Priority**	**Type**
FR-01	Access Control	The system should allow a member to scan a QR code using a mobile device or kiosk for identification.	High	Mandatory
FR-02	Access Control	The system should validate the member's subscription status in real-time upon a scan attempt.	High	Mandatory
FR-03	Access Control	The system should authenticate the user's	High	Mandatory

		identity based on the scanned QR code data.		
FR-04	Access Control	Upon successful authentication and validation, the system should send a signal to unlock the entrance door.	High	Mandatory
FR-05	Access Control	The system should record a timestamp with member ID, event type (IN/OUT), and location for every entry and exit.	High	Mandatory
FR-06	Access Control	The system should prevent duplicate "IN" events for a member who is already logged inside.	Medium	Mandatory
FR-07	Access Control	The system should provide instructive error messages for failed scans, invalid subscriptions, or	Medium	Mandatory

		authentication failures.		
FR-08	Access Control	The system should allow staff members to enter/leave the facility, logging their access.	Medium	Mandatory
FR-09	Membership	The system should allow a member to purchase a new subscription plan via the web or mobile app.	High	Mandatory
FR-10	Membership	The system should integrate with a payment gateway to process subscription payments.	High	Mandatory
FR-11	Membership	Upon successful payment, the system should automatically activate the member's subscription and update their record.	High	Mandatory

FR-12	Membership	The system should send a confirmation notification (e.g., email) to the member after a successful purchase.	Medium	Mandatory
FR-13	Scheduling	The system should allow members to view available time slots and instructors for appointments.	Medium	Mandatory
FR-14	Scheduling	The system should allow a member to book an appointment with a selected instructor for an available time slot.	Medium	Mandatory
FR-15	Scheduling	The system should verify a member's subscription entitlements before allowing an appointment to be booked.	Medium	Mandatory
FR-16	Scheduling	The system should send	Medium	Mandatory

		notifications to both the member and the instructor upon appointment creation.		
FR-17	Scheduling	The system should allow members and instructors to modify or reschedule existing workout appointments.	Medium	Mandatory
FR-18	Scheduling	The system should check for scheduling conflicts (e.g., trainer availability) before confirming any modification.	Medium	Mandatory
FR-19	Progress Tracking	The system should allow a member to mark a scheduled workout or routine as completed.	Medium	Mandatory
FR-20	Progress Tracking	The system should record a completion timestamp	Medium	Mandatory

		when a workout is marked as done.		
FR-21	Progress Tracking	The system should recalculate and update the member's progress statistics upon workout completion.	Medium	Mandatory
FR-22	Progress Tracking	The system should display a dashboard to the member with historical workout data, trends, and visualizations.	Medium	Mandatory
FR-23	Administration	The system should provide an admin console for managing facilities, user roles, pricing plans, and system settings.	Low	Mandatory
FR-24	Reporting	The system should allow managers to generate and export detailed reports (e.g.,	Low	Mandatory

		attendance, revenue) in PDF/CSV formats.		
FR-25	Security	The system should validate user roles and permissions before allowing access to any functionality.	High	Mandatory
FR-26	Analytics	The system should provide an insights dashboard for managers showing trends, occupancy patterns, and recommended actions.	Low	Optional
FR-27	Data Management	The system should store data locally and sync later if a network connection is unavailable (offline capability).	Medium	Optional
FR-28	Audit	The system should log all user and administrative	Low	Mandatory

		actions for audit purposes.		

Requirements analysis – Non-Functional

Non-functional requirements define the standards and attributes that shape how a system performs, rather than the specific tasks it carries out. These requirements play a critical role in ensuring the system operates smoothly, efficiently, and remains dependable over time.

They typically emphasize:

1. Performance – How fast, responsive, and efficient the system is under various conditions.

2. Reliability & Availability – The system's ability to function consistently and remain accessible when needed.

3. Scalability & Maintainability – How well the system adapts to growth and how easily it can be updated or improved.

ID	**Category**	**Requirement Description**	**Priority**	**Type**
NFR-01	Security	The system shall use secure, signed, or one-time-use QR codes to prevent replay attacks.	High	Mandatory
NFR-02	Security	The system shall implement rate-limiting on authentication attempts to prevent brute-force attacks.	High	Mandatory

NFR-03	Security	All sensitive data (passwords, payment info) shall be encrypted in transit (TLS 1.2+) and at rest (AES-256).	High	Mandatory
NFR-04	Performance	The system shall respond to a QR scan and unlock door signal in under 2 seconds to prevent entrance delays.	High	Mandatory
NFR-05	Performance	The reporting module should generate standard reports for a 30-day period in under 10 seconds.	Medium	Optional
NFR-06	Reliability	The core access control system shall have 99.9% uptime during gym operating hours.	High	Mandatory

NFR-07	Reliability	The system should gracefully handle failure of a secondary service (e.g., if email service is down, queue notifications).	Medium	Optional
NFR-08	Usability	The system shall provide clear, accessible, and multi-language user interface messages and instructions.	Medium	Mandatory
NFR-09	Usability	The member mobile app should be learnable for a new user to perform basic tasks (check-in, view schedule) within 5 minutes.	Medium	Optional
NFR-10	Compatibility	The member mobile app shall be compatible with iOS (version XX and above) and Android (version YY and above).	High	Mandatory

NFR-11	Compatibility	The system shall integrate with standard payment gateways (e.g., Stripe, PayPal) using common APIs.	High	Mandatory
NFR-12	Scalability	The system architecture should support a 20% increase in member count without a loss in performance.	Low	Optional
NFR-13	Maintainability	The system should be modular, allowing individual components (e.g., billing, auth) to be updated independently.	Low	Optional
NFR-14	Data Integrity	The system shall validate event ordering to prevent illogical records (e.g., two consecutive "IN" events).	Medium	Mandatory

Feasibility Study

Technical Feasibility

- Assessment: The proposed system is Technically Feasible.
- Technology Availability: All core technologies required are mature, readily available, and well-understood.

- Front-end: Mobile apps (iOS/Android) can be developed with frameworks like React Native or Flutter. Web interfaces for admin and members can use standard JavaScript frameworks (React, Angular).
- Back-end: Standard cloud providers (AWS, Azure, Google Cloud) offer all necessary services (databases, serverless functions, authentication services).
- Hardware: QR scanners, tablets for kiosks, and electronic door locks are commercial off-the-shelf (COTS) products with standard APIs for integration.

- Integration Complexity: The system requires integration between several modules (auth, billing, scheduling) and third-party services (payment gateways like Stripe). This presents a medium level of complexity but is a common practice with well-documented APIs and can be managed by a competent development team.
- Technical Skills Required: Development requires standard skills in full-stack web and mobile development, API design, and database management. These skills are widely available in the market. Maintenance post-launch requires a simpler skill set, manageable by a system administrator or a managed service provider.
- Scalability: The proposed cloud-based architecture is inherently scalable. Database and server resources can be increased to handle a growing member base without significant architectural changes.

- Conclusion: No insurmountable technical barriers exist. The project can be developed with existing technology and skills.

Financial Feasibility

- Assessment: The proposed system is Financially Feasible with a positive expected Return on Investment (ROI).

- Costs:
  - Development Costs: One-time cost for custom software development. This is the largest upfront investment.
  - Hardware Costs: One-time purchase of QR scanners, and electronic door locks.
  - Software Costs: Recurring subscriptions for cloud hosting, payment gateway fees, and potential third-party API services.
  - Maintenance & Support: Recurring annual cost for updates, security patches, and technical support (could be ~15-20% of initial development cost per year).

- Benefits & Cost Savings (ROI Justification):

- Reduced Revenue Leakage: Automated subscription validation at the door (FR-02, FR-03) will directly prevent lost income from expired or non-existent memberships. This is a direct financial gain.
- Operational Efficiency: Significant reduction in administrative time spent on manual registration, attendance tracking, and scheduling (Current UC-01, UC-02, UC-03). Staff can be reallocated to higher-value tasks like member service.
- Increased Revenue Streams: Streamlined trainer booking (FR-14) can increase personal training sales. Member retention is likely to improve due to the enhanced experience (progress tracking, convenience), reducing member churn.

- Improved Decision-Making: Data-driven insights from reports (FR-24) can optimize marketing spend, class schedules, and resource allocation, leading to better financial outcomes.
- Conclusion: While there are significant upfront costs, the long-term benefits of increased revenue, reduced losses, and operational savings present a strong business case for a positive ROI, especially when implementing the core modules (BSO 1 & 2) first.

Operational Feasibility

- Assessment: The proposed system is Operationally Feasible and will significantly improve workflows.
- Alignment with Current Procedures: The system is designed to replace inefficient manual processes (paper cards, sign-in books) with streamlined digital ones. It directly addresses the identified pain points.
- Usability and Training:

➢ Members: The interface for scanning QR codes and booking appointments is intuitive and similar to other modern apps. The learning curve is low.

➢ Staff: The admin and instructor dashboards will require training. However, they simplify complex tasks (scheduling, member lookups), so adoption resistance should be low once the benefits are demonstrated.

- Impact on Workflow: The impact is profoundly positive.
- Front Desk: Eliminates manual registration and attendance tracking. Staff become enablers rather than gatekeepers.
- Instructors: Frees them from administrative scheduling tasks, allowing them to focus on coaching.

- Management: Provides real-time visibility into gym operations and member engagement, which is currently non-existent.
- Conclusion: The system will integrate well into the gym's operations. It solves more problems than it creates and is designed with end-user usability in mind.

Organizational Feasibility

- Assessment: The proposed system is Organizationally Feasible, but requires change management.
- Management Support: As outlined in the current system, "Top Management" oversees operations. Their buy-in is critical. The feasibility study and clear ROI presentation are essential to secure their sponsorship.
- User Acceptance: This is the primary potential hurdle.
- Members are likely to welcome the modern convenience, similar to systems used in other industries.
- Staff/Instructors may initially resist due to fear of change or perceived deskilling. This must be managed through:

➢ Clear Communication: Explain the "why" – the system will make their jobs easier, not replace them.

➢ Comprehensive Training: Ensure all staff are confident using the new tools.

➢ Phased Implementation: Start with access control (BSO 1), which has obvious benefits, before rolling out more complex modules.

- Organizational Culture: The shift from a low-tech, manual environment to a data-driven, automated one represents a significant cultural change. Leadership must champion this change to create a culture that embraces efficiency and data.
- Conclusion: The project is viable from an organizational perspective, but its success is contingent on effective change management strategies, strong leadership support, and thorough user training to ensure smooth adoption.

Overall Conclusion

The proposed Gym Management System is feasible across all four evaluated dimensions. It is technically sound, financially justified, operationally beneficial, and organizationally viable with proper management. It is recommended to proceed with the project, adopting a phased implementation approach (e.g., starting with BSO 1: Essential Digitalization & Access Control) to manage risk, budget, and organizational change effectively.

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